

Calf pain

with TAMIM KHANBHAI, MATT HISLOP and JEFF BOYLE

Kompany remains sidelined with a calf injury, sustained after just seven minutes of City's goalless draw with Dynamo Kiev one month ago. Remarkably, it is the 14th time he has suffered a problem with his calf since joining the club in 2008—with five of them occurring this season—and his 32nd injury overall during the past eight years.

PhysioRoom, 12 April 2016

Calf pain is common and if not managed appropriately it can persist for months and cause frustration for both athlete and clinician. Both acute and chronic calf pain can stem from injury to the calf muscle.

ANATOMY

The anatomy of the posterior compartment of the leg (Fig. 39.1) will be covered here. The anterior and lateral aspects have been covered in Chapter 38.

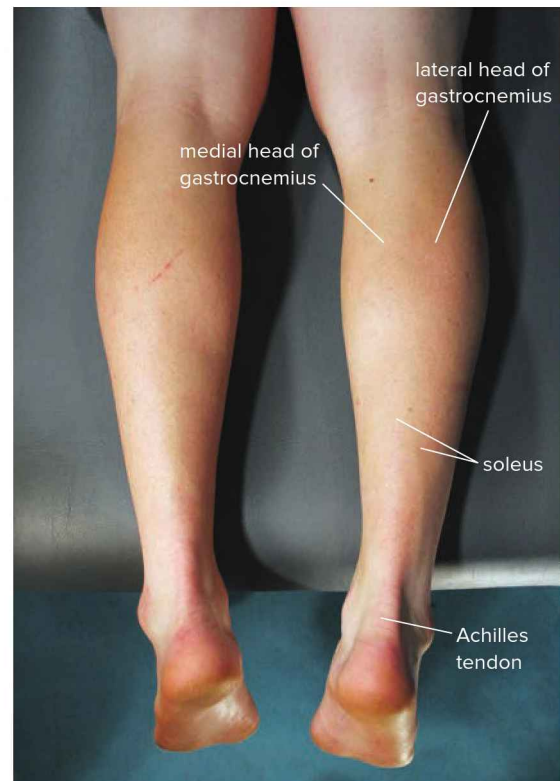
The muscles in the posterior compartment of the leg comprise a superficial and deep group. The superficial group is composed of the triceps surae (gastrocnemius and soleus) and plantaris muscle. These form the bulk of the calf muscle. The deep group is composed of the tibialis posterior, flexor digitorum longus, flexor hallucis longus and popliteus muscles.

Gastrocnemius has two heads; the medial and larger head is attached to the posterior aspect of the medial epicondyle of the femur and the lateral head to the lateral femoral condyle (Fig. 39.1b). The muscle extends down to the mid-calf, following which it inserts into a broad aponeurosis which receives the tendon of the soleus to form the tendon of Achilles (TA). The two heads are overlaid by the tendon of biceps femoris and the common peroneal nerve laterally and the semimembranosus muscle medially. This latter area between the medial head and semimembranosus is the location of the 'Baker's cyst' (Chapter 37).

Soleus is a large flat broad muscle, deep to the gastrocnemius (Fig. 39.1c). It arises from a number of locations; the posterior aspect of the head and proximal shaft of the fibula, the soleal line (prominent ridge posterior tibia), medial aspect of the tibia and a fibrous band between the tibia and fibula. The muscle joins with

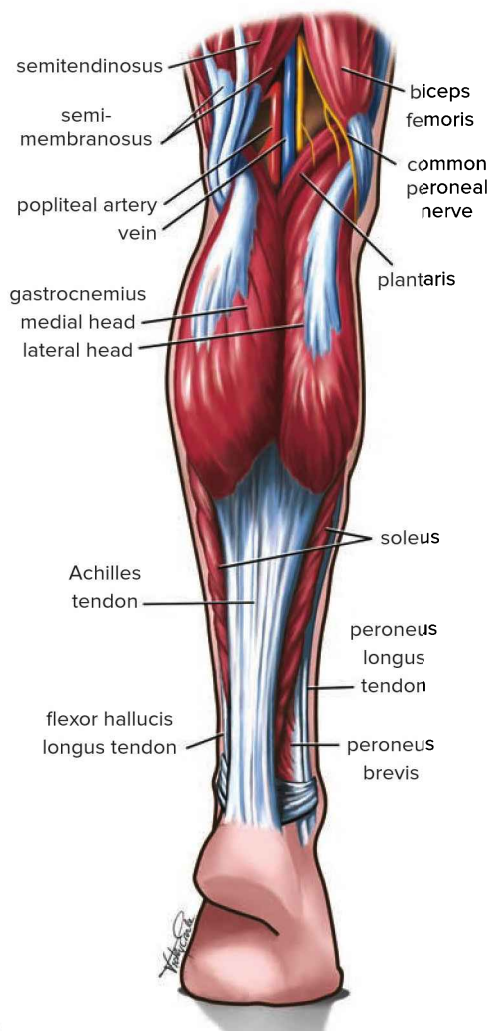
the tendon of gastrocnemius and together they form the TA, inserting into the calcaneal tuberosity.

Both the gastrocnemius and soleus muscles in the calf have been noted to have intramuscular tendons. In the

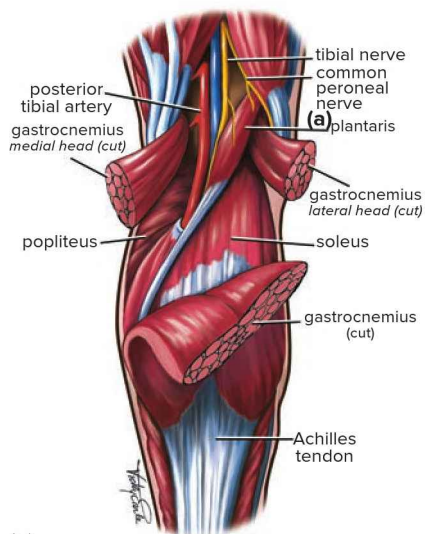


(a)

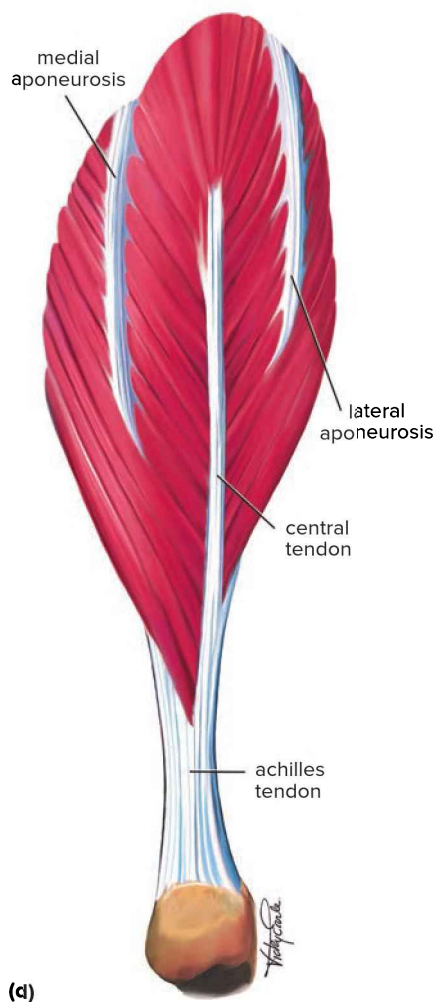
Figure 39.1 Anatomy of the calf (a) Surface anatomy



(b)



(c)



(d)

Figure 39.1 (cont.) (b) Superficial calf muscles (c) Removal of the gastrocnemius showing the underlying soleus and plantaris muscles (d) The intramuscular tendinous structures within the soleus muscle

gastrocnemius muscle, separate proximal medial and lateral tendons arise from the posterior aspects of the medial and lateral femoral condyles respectively.

There are three tendinous structures within the soleus muscle—medial and lateral ‘aponeuroses’ proximally, and a distal central tendon.¹ The medial and lateral aponeuroses (Fig. 39.1d) arise from the medial (tibial) and lateral (fibular) origins of the soleal epimysium and descend inferiorly, penetrating deep into the medial and lateral aspects of the soleal muscle belly, whilst simultaneously curving towards the midline of the muscle belly and distal central tendon. As they progress distally, the aponeuroses decrease in size until they can no longer be visualised. This most distal point is from where the lowest muscle

fibres arise. As these aponeuroses are a continuation of the epimysium covering the soleus and act as rigid fibrous 'struts' from which the proximal soleus muscle fibres gain origin, they can indeed be considered as true intramuscular tendons from an anatomical, functional and pathologic perspective, albeit demonstrating a flattened and 'aponeurotic' morphology.

The central tendon (Fig. 39.1d) arises from the anterior aponeurosis and consequently represents the inferior muscular attachment; that is to say, the insertion of the soleal muscle fibres. This eventually forms the soleal contribution of the Achilles tendon, typically fusing along the most central aspect of the tendon, although it occasionally does so along the medial or lateral aspect.

CLINICAL APPROACH

Calf injuries are common, particularly in ball sports. For example, a large study of injuries in professional football found that 13% of all injuries were located in the calf.² Of these, 61% resulted in more than a week of time loss and 13% took more than 28 days to recover. The re-injury rate was also high at 13%.

The medial head of the gastrocnemius is the most common site of calf injury. In an MRI study of calf injuries by Koulouris and colleagues,³ a total of 79 separate sites of strain injury were identified (39 solitary, 20 dual) in 59 patients. Of the 39 isolated strains, injury to the gastrocnemius was most common (49%), preferentially involving the medial head in 18 cases and the lateral head in one case. The soleus was also commonly involved (46%), with two cases (5%) of distal avulsions of the plantaris. Of the 20 dual injuries, a combination of gastrocnemius injury with soleus injury was the most frequent finding (60%). Dual injuries of both heads of the gastrocnemius muscles were demonstrated in four cases (20%), with the soleus and tibialis posterior injured in three cases (15%). A combination of soleus and flexor hallucis longus injury was seen in one case (5%).

Most calf muscle strains occur at the musculotendinous junction (MTJ)—either at the intramuscular tendon or the distal MTJ. Balius and colleagues found that musculotendinous junction strains involved all three of the soleus intramuscular tendons.¹ Of 31 strains considered musculotendinous, 14 (45%) were located at the medial aponeurosis, 10 (32%) at the central tendon and seven (23%) involved the lateral aponeurosis.

A sudden burst of acceleration, such as stretching to play a ball in squash or tennis, may precipitate injury. The classic mechanism is seen in middle-aged tennis players, normally over 40 years of age, with a sudden onset of acceleration, where the calf muscle is stretched eccentrically when the knee is extended and the foot in a dorsiflexed position, such as when overstretching for a ball. This results in immediate pain, tenderness (medial belly of

gastrocnemius above the musculotendinous junction), a palpable defect and swelling. This is sometimes referred to as 'tennis leg'.

The calf region is also a common site of contusion caused through contact with playing equipment or another player. Muscle strains and contusions are acute injuries that present with typical histories that are usually easily distinguishable.

PRACTICE PEARL

The possibility of referred pain from the lumbar spine, neural or myofascial structures should always be considered. This is sometimes the underlying basis of calf strains, especially in the older athlete.⁴

The calf is the most common site in the body for muscle cramps (Chapter 3). Cramps may occur at rest, during or after exercise. Cramps are not specific to environmental conditions such as exercise in the cold or heat. They can occur in acclimatised and conditioned athletes. Cramps probably result from alterations in spinal neural reflex activity activated by fatigue in susceptible individuals.

The calf is also a common site of the phenomenon known as delayed onset muscle soreness (Chapter 4). This may occur after the first training session following a lay-off or a period of relative rest, or when excessive eccentric muscle contractions are performed, for example, during plyometric training. Lateral calf pain may be due to a direct blow, referred pain from the superior tibiofibular joint, peroneal muscle strain or fibular stress fracture. The causes of calf pain are listed in Table 39.1.

A variety of contributing factors can predispose an athlete to calf injury. These include:

- foot biomechanics (excessive pronation/supination)
- pelvic biomechanics (e.g. internal rotation femur)
- muscle imbalances (quadriceps, hamstrings, ankle everters and inverters)
- stiff ankle joint (decreased dorsiflexion)
- poor jumping and landing technique
- change in sporting environment (terrain, weather)
- poor nutrition and hydration
- change in sporting activity (cycling and swimming compared with running)
- change in footwear
- reduced muscle power, endurance and muscle fatigue
- lack of conditioning.

History

The most important aspect of the history is the description of the onset of pain. A sudden onset of a tearing or

Table 39.1 Causes of calf pain

Common	Less common	Not to be missed
Muscle strains • Gastrocnemius • Soleus • Plantaris Muscle contusion • Gastrocnemius Muscle cramp Delayed onset muscle soreness	Superficial posterior compartment syndrome Deep posterior compartment syndrome (Chapter 38) Referred pain • Lumbar spine • Myofascial structures • Superior tibiofibular joint • Knee (Baker's cyst, posterior cruciate ligament, posterior capsular sprain) Vascular entrapment • Popliteal artery • Endofibrosis of external iliac artery Nerve entrapment • Tibial • Sural Stress fracture of the fibula Stress fracture of the posterior cortex of the tibia (Chapter 38) Varicose veins (e.g. superficial thrombosis)	Deep venous thrombosis

popping sensation in the calf is diagnostic of calf muscle strains. The patient is usually able to localise the site of the tear. The degree of disability, both immediately after the injury and subsequently, is a guide to the severity of the tear.

In athletes with chronic mild calf pain, a history of a previous acute injury may be significant and is a risk factor for calf injury in marathon runners and soccer referees.^{5, 6} The practitioner should ask about the treatment and rehabilitation of the previous injury because inadequate rehabilitation will have an impact on recurrence rates.

Information about the training load, and especially running load, can be very helpful. Many runners now use GPS devices or mobile applications that provide extensive information about running distances, speed and total volume per week. This information can be very valuable.

A history of low back pain may be a clue to referred pain. The practitioner should also be alert to the possibility of referred pain if the calf pain is variable rather than constant. If there is no obvious precipitating cause of calf pain, the possibility of deep venous thrombosis after long car journeys, long flights or recent surgery needs to be explored.

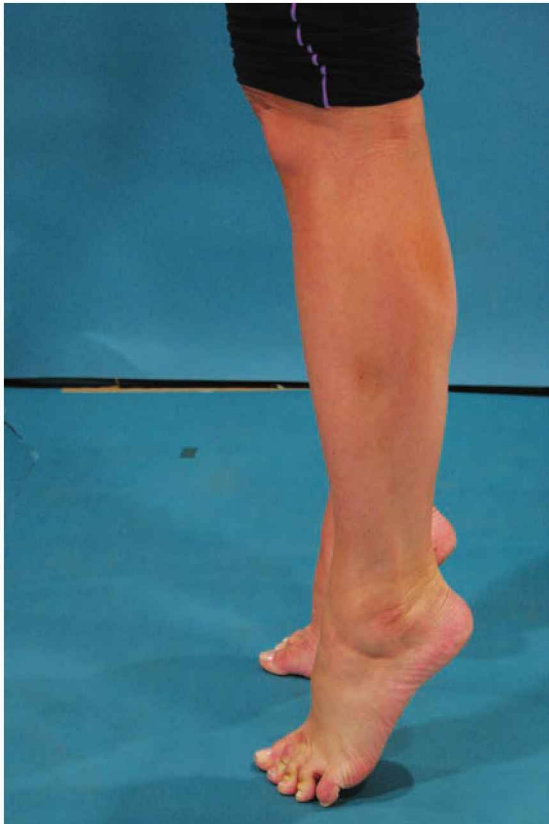
Claudicant pain that comes on with exertion and then disappears may indicate proximal vascular occlusion by atheroma or entrapment (e.g. femoral or iliac artery stenosis) or exertional compartment syndrome. A ruptured Baker's cyst can also cause acute calf pain and swelling and should be considered as a differential diagnosis (refer to Chapter 37).

Physical examination

The aims of examination in the patient with acute calf strain are to determine the site and the severity of the injury as well as to detect any predisposing factors such as chronic calf tightness. Examination of a patient with chronic or intermittent calf pain requires not only palpation of muscles, but also assessment of possible neural contributors, ankle reflexes and examination of the lumbar spine. It is important to palpate the entire length and width of the muscle bellies and their associated aponeuroses for areas of muscle and fascial tightness and thickenings that may predispose to injury. Due to its deep position, the soleus can be difficult to palpate.

Clinical assessment should include both static and dynamic tests and should be performed in both knee flexion and extension. Comparing calf girth between left and right is also important. Foot alignment and biomechanics should also be considered.

1. Observation
 - a. standing (foot posture, calf girth, swelling)
 - b. walking (on toes, on heels)
 - c. prone
2. Active ankle movements
 - a. plantarflexion/dorsiflexion (standing) (Fig. 39.2a)
 - b. plantarflexion/dorsiflexion (prone)
3. Passive ankle movements
 - a. dorsiflexion (knee flexion) (Fig. 39.2b)
 - b. dorsiflexion (knee extension) (Fig. 39.2c)



(a)



(b)



(c)



(d)

Figure 39.2 Examination of the patient with calf pain (a) Active movement—plantarflexion/dorsiflexion (standing). The functional competence can be assessed during a bilateral or unilateral heel raise-and-drop until pain is reproduced (b) Passive movement—ankle dorsiflexion (knee flexion). Examine with the knee flexed and add overpressure (c) Passive movement—ankle dorsiflexion (knee extension). Examine with the knee extended and add overpressure (Homan's sign) (d) Stretch—gastrocnemius (back leg). Examine with the knee in full extension and the heel on the ground

c. muscle stretch

- i. gastrocnemius (Fig. 39.2d)
- ii. soleus (Fig. 39.2e)

4. Resisted movements

- a. dorsiflexion

5. Functional tests

- a. hop (hop for distance both straight leg and bent leg)
- b. jump (height, shock absorption)
- c. run
- d. calf raises to fatigue (both straight leg and bent leg)

6. Palpation

- a. gastrocnemius (Fig. 39.2f) (compare medial and lateral heads)



(e)



(f)



(g)



(h)

Figure 39.2 (cont.) (e) Stretch—soleus (lunge test). The patient should flex the knee so that it passes vertically over the third toe to prevent excessive pronation. Record the range of motion and compare both sides (f) Palpation. The patient should actively contract and relax the muscles and the ankle should be moved passively through dorsiflexion and plantarflexion during palpation. The gastrocnemius may be palpated in the relaxed position by placing the knee in flexion and the ankle in plantarflexion. Feel for swelling and defects in muscle or tendon tissue (g) Special test—calf squeeze test. The calf is squeezed. If no ankle plantarflexion occurs, there has been a complete tear of the Achilles tendon or musculotendinous junction (h) Special test—neurodynamic test (slump test)

- b. soleus
 - c. posterior knee
 - d. superior tibiofibular joint
 - e. Achilles tendon
7. Special tests
- a. Calf squeeze test—for Achilles tendon rupture (Fig. 39.2g) (Chapter 40)
 - b. Homan's sign—forced dorsiflexion in knee flexion may elicit pain in the presence of deep venous thrombosis (Fig. 39.2c). However, this is not a sensitive or specific test⁷
 - c. neurodynamic test—slump test (Fig. 39.2h)
 - d. gluteal trigger points
 - e. lumbar spine (Chapter 29)
 - f. biomechanical examination (Chapter 8)
 - g. ankle jerk reflex
 - h. popliteal artery auscultation after provocative manoeuvres.

Key outcome measures

There are no specific PROMs for calf strains. Generic measures such as a patient-specific functional scale or a numeric pain rating scale can be used to monitor progress (see Chapter 16).

Investigations

Ultrasound or MRI may be useful for evaluating a calf injury. These imaging modalities can localise pathology, identify presence of haematoma and scar tissue, and differentiate between a muscle strain and a contusion if not clinically evident. MRI is preferred because of its ability to identify intramuscular tendon injuries. If deep venous thrombosis is suspected, a Doppler scan may be required.

GASTROCNEMIUS MUSCLE STRAINS

Acute strain

Acute strain of the gastrocnemius muscle occurs typically when the athlete attempts to accelerate by extending the knee from a stationary position with the ankle in dorsiflexion, or when lunging forward, such as while playing tennis or squash. Sudden eccentric overstretch, such as when an athlete runs onto a kerb and the ankle drops suddenly into dorsiflexion, is another common mechanism.

The medial head of the gastrocnemius has a greater proximal attachment and a longer distal insertion into the Achilles tendon than the lateral head.³ Therefore the medial head may have a greater capacity for force generation and may be more susceptible to injury compared to the lateral head.

History

The exact moment of injury was caught on video in the case of a famous Australian batsman whose gastrocnemius strain occurred when his entire body weight was over his foot on the injured side with the centre of mass well in front of the leg.⁸ The gastrocnemius muscle-tendon complex was at close to maximal length, and the muscle-tendon length was also constant at the time. Therefore, the injury probably occurred just as the muscle-tendon complex was moving from an eccentric to an isometric phase.

The athlete often complains of an acute, stabbing or tearing sensation, usually either in the medial belly of the gastrocnemius or at the musculotendinous junction.

Physical examination

Examination reveals tenderness at the site of a muscle strain and, in complete muscle tears, there may be a palpable defect. The athlete will walk with an antalgic gait. There may be skin changes such as bruising or discoloration and swelling. Assessing range of motion reveals pain on passive and active dorsiflexion (Fig. 39.2d), and on resisted plantarflexion with the knee extended. Assessment of the Achilles tendon and other structures in the calf and ankle are important.

If required, the functional competence of the injured muscle is assessed by asking the patient to perform a bilateral heel raise. If necessary, a unilateral heel raise, heel drop or hop may be used to reproduce the pain and to assess for weakness. Measuring the calf girth, comparing one side to the other in centimetres and a lunge test, when appropriate (not if examining the athlete acutely), can be useful objective markers during rehabilitation, and especially if there are baseline measurements for comparison.

Chronic gastrocnemius muscle strain may occur as an overuse injury or following inadequate rehabilitation of an acute injury. This results in disorganised, weak scar tissue that can predispose to further injury and compensatory movement patterns leading to muscle imbalances. Careful palpation of the muscle can reveal these tissue thickenings.

Treatment

Rehabilitation consists of a progressive program starting from the acute phase through to the athlete returning to sport. It is tailored to the individual, rather than a fixed time frame. Some athletes will recover faster than others, depending on the risk factors of the injury, presence of scar tissue, haematoma, previous injuries and the sport they need to return to. The aim is to return the athlete, at a minimum, to their pre-injury level, but ideally to use

this opportunity to correct other issues so they are in a better position than their pre-injury level. There are five stages to rehabilitation of gastrocnemius strains:

- Stage 1: Acute phase
- Stage 2: Improve ROM
- Stage 3: Power and endurance
- Stage 4: Coordination, agility, sport skills
- Stage 5: Return to sport.

Stage 1: Acute phase

Initial treatment of gastrocnemius strains aims to reduce pain, swelling and further injury. The initial approach is POLICE (protect, optimal loading, ice, compression, elevation)⁹ and analgesics may be used if required, avoiding nonsteroidal anti-inflammatory drugs for the first 72 hours.^{10, 11} Ice compression units may assist with this process, for 15-20 minutes every hour. Crutches may be necessary if the athlete is unable to bear weight, but should only be used for short periods as early mobilisation allows muscle fibres to regenerate and organise more effectively.¹²

A 6 mm heel raise may be used (on both the injured and uninjured side) to initially reduce calf length and off-load calf structures. There is no good quality evidence for electrotherapeutic modalities such as ultrasound therapy, but this can be used if the treating team is keen to incorporate this.

This stage can be achieved within a couple of days but can be longer depending on the severity of the injury. If the symptoms have failed to settle within 3-5 days, reassess and investigate further if required.

Stage 2: Improve ROM +/- flexibility

Once the pain and swelling has settled, ankle and knee active range of motion exercises should commence and exercises in a swimming pool can aid recovery.

Gentle stretching of the gastrocnemius to the level of a feeling of tightness (Fig. 39.3) can begin soon after the injury. Stretching is important to lengthen the muscle, which improves its elasticity and distends scar tissue appropriately.¹² Following this, unloaded isometric calf exercise should be incorporated.

Stage 3: Power and endurance

There are multiple facets to this stage. Within 10 days the scar tissue will have the same tensile strength as normal muscle and can be subjected to increased loading.¹³ Progression from active ROM to strengthening of the plantarflexors can commence. This involves a progression of exercises starting with resistance band strengthening to concentric bilateral calf raises, unilateral calf raises and, finally, eccentric calf lowering. Weights can be

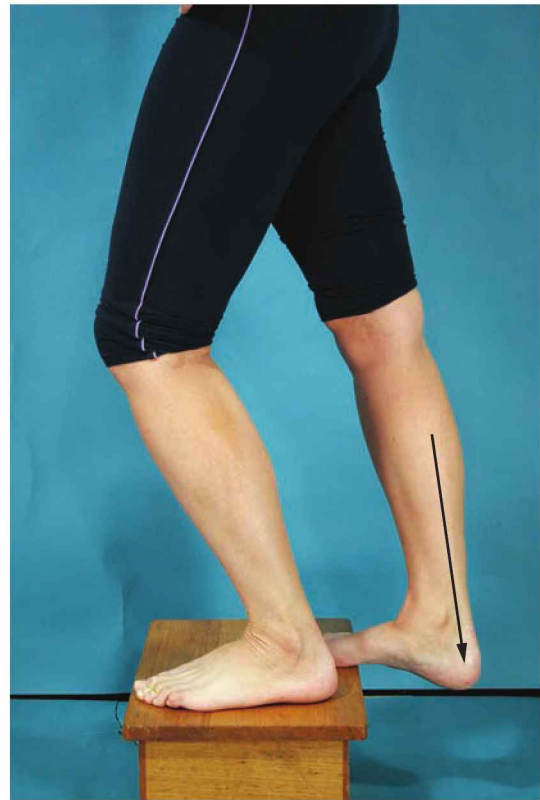
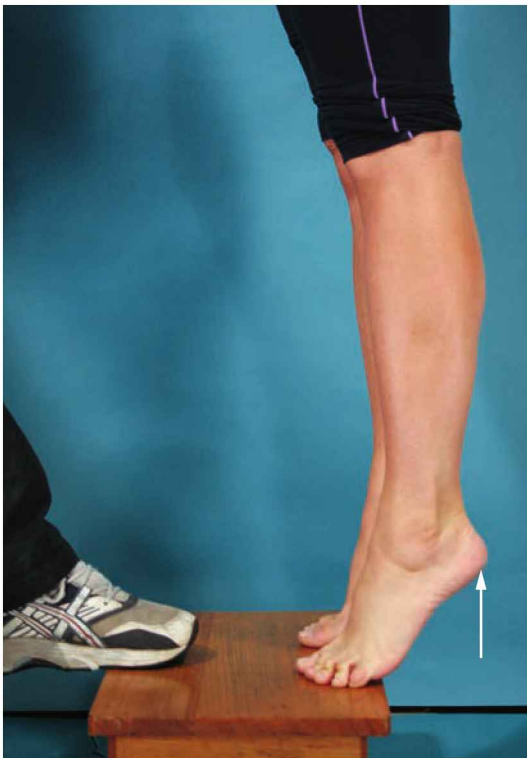


Figure 39.3 Stretching exercise for the right gastrocnemius muscle with the knee in full extension. This can be performed over a step or with the foot placed against a wall to increase the stretch

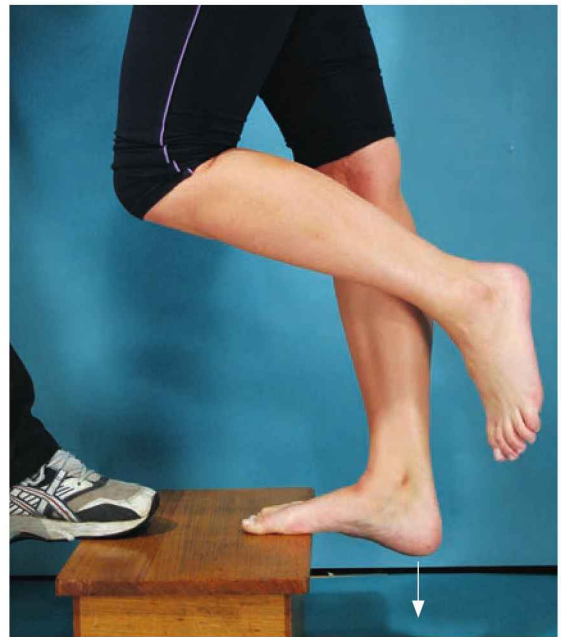
added gradually at each stage (Fig. 39.4). Strengthening routines should begin with open chain then closed chain exercises.

Prior to rehabilitation, ensure the muscle has been warmed up appropriately. Rehabilitation should concentrate on both gastrocnemius and soleus by varying knee extension and flexion. Stretching exercises will restore normal muscle length and should be held for at least 15 seconds. A systematic review of calf stretching on ankle dorsiflexion range of motion found an improvement of up to 2-3°,¹⁴ but it is unknown whether this is clinically relevant. Measuring the lunge test is a useful indicator of posterior calf flexibility as a score <13 cm has been associated with lower limb injuries.¹⁵

Treatment may also include transverse friction and longitudinal gliding soft tissue therapy at the site of any excessive scar tissue and along the entire musculotendinous unit. When appropriate, sustained myofascial tension (Fig. 39.5) or digital ischaemic pressure can be applied to areas of increased tone.



(a)



(c)

Figure 39.4 Strengthening exercises for the gastrocnemius muscle (a) Bilateral calf raise (b) Unilateral calf raise (c) Eccentric lowering



(b)

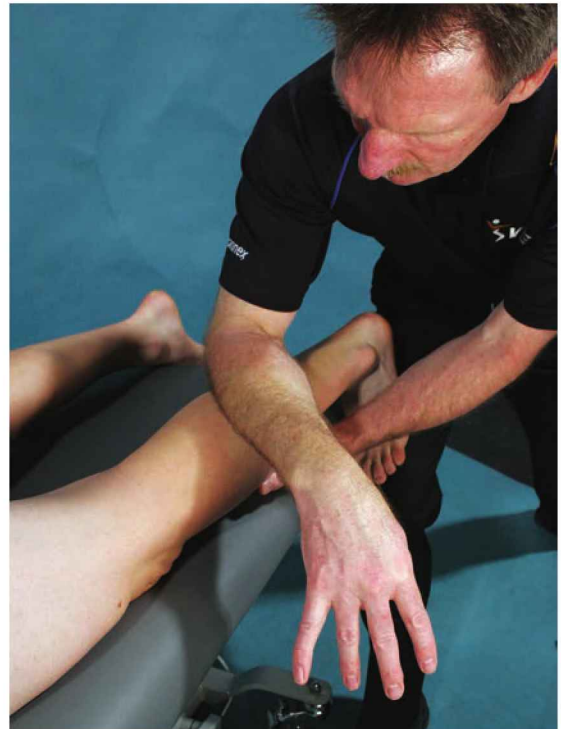


Figure 39.5 Soft tissue therapy—sustained myofascial tension to the muscle belly of the gastrocnemius

For favourable long-term results, treatment should address predisposing factors such as biomechanical factors, or risk factors that preceded the injury, such as calf muscle tightness, lumbopelvic dysfunction and knee or ankle-related abnormalities.

Normal, full weight-bearing gait should be achieved, following which proprioceptive exercises can be started. This begins with balancing on the injured leg, progressing through to a wobble board or BOSU® ball and incorporating functional movements like jumping onto or landing off the board once the strength in the calf is appropriate.

Increased amounts of cardiovascular exercise should be undertaken to maintain fitness levels. Low-impact cross training such as stationary cycling or swimming can be commenced as soon as pain allows. When active weight-bearing muscle contraction is pain-free, a cross trainer can also be used prior to running.

Stage 4: Coordination, agility, sport skills

By this stage, the strength on the injured side should be comparable to the non-injured leg. The aim now is to get the player ready for return to play. This involves continuing the gains made in the previous stage, such as strength work and proprioception. Now there needs to be a supervised, graduated running program, which should include sprint work, change of direction and change of pace. If the sport is football, for example, then during this stage ball work should be introduced, starting with short kicks and progressing to longer kicking drills. Athletes should undergo a supervised, graduated return-to-sport program.

Stage 5: Return to sport

This stage is often overlooked but is integral in a rehabilitation program (Chapter 19). Return to sport (RTP) needs a multidisciplinary approach, working with the coaching staff, informing and educating them about the injury. Often, coaches will feel that once a medical team hands a player back to them, they are ready to play. In fact, nothing in rehabilitation can replace or replicate the training and matches of team sport.

The aim should be to have a period of training with the team, the length of which is decided by the medical team and depends on factors such as the time out with that injury. Clearly if a player has been out for 1 week, they need less training time than if they are out for a period of 4–8 weeks. By this stage, strength, flexibility and proprioception should be equal to the uninjured site.

There is little evidence that a repeat MRI scan needs to be done to RTP as MRIs still show abnormalities 6 weeks after players had returned to play in Australian rules football.¹⁶ The number of minutes the player participates in the sport should be lower than usual when the player first returns. This stage needs to be tailored to the individual,

the injury and the sport. Re-injuries often occur because the importance of this stage is ignored.

PRACTICE PEARL

When a 'tennis leg' occurs in an older athlete it can take a few months before the individual is ready to return to sport. In such cases there is often fibrotic tissue between the gastrocnemius and soleus muscles, seen on follow-up ultrasound. This fibrotic tissue may increase the risk of recurrence. One of the aims with any treatment is to achieve optimal rehabilitation and to prevent the formation of fibrotic tissue or scarring.

Operative intervention

Surgery is rarely required for gastrocnemius tears. However, if there is a complete rupture of the medial gastrocnemius head or a large disruption of muscle or intramuscular tendon fibres then a surgical opinion should be sought. Likewise, a failure of conservative treatment or prolonged pain (4–6 months) with a contracture warrants a surgical opinion.¹³ Surgical intervention is only indicated when the presentation is associated with acute compartment syndrome.¹⁷ Reconstruction of the medial head of gastrocnemius has limited evidence, with few case reports, a small case series and no long-term follow up data.¹⁸ It is thus reserved for the rarer cases of this type of injury.

SOLEUS MUSCLE STRAINS

The soleus muscle contains a high proportion of type I, slow-twitch muscle fibres, unlike the gastrocnemius, which contains a higher proportion of type II, fast-twitch muscle fibres. This allows the soleus muscle to have a significant role in posture control, especially in standing.

Soleus muscle strains occurring at the perimeter of the muscle, and not centred at the tendons, can be classified as myofascial (epimysial) strains. These may be at the myofascial junction of the soleus with the gastrocnemius (posterior myofascial), or at the junction between soleus with the deep posterior compartment of the leg (anterior myofascial). Musculotendinous junction strains can involve the distal intramuscular tendon, or the proximal medial and lateral aponeuroses (Fig. 39.1d).

Epidemiology

Strains of the soleus muscle were not thought to be common in the elite athlete,¹⁷ but Koulouris who reviewed MRI findings rather than purely ultrasound found large numbers of soleus strains, and combination strains with gastrocnemius and other deep crural muscles to be common.³ Of the 18 isolated soleus strains reported by Koulouris, 11 were described as myofascial, with the other seven as musculotendinous junction injuries, all medial.

Table 39.2 Distribution of soleus muscle strains

Myofascial vs musculotendinous strains			
	Myofascial		Musculotendinous
Koulouris et al ³	11		7
Balius et al ¹	24		31
Distribution of musculotendinous strains			
	Medial	Lateral	Central
Koulouris et al ³	7	0	0
Balius et al ¹	14	7	10

Balius reported a series of 55 cases of soleus muscle strains, of which 31 strains were considered musculotendinous and 24 were myofascial.¹ The distribution of the soleus strains in the two studies is shown in Table 39.2.

MRI demonstrates the differences between the myofascial (epimysial) strains (Fig. 39.6) and the more serious musculotendinous tear involving damage to the aponeurosis (Fig. 39.7).

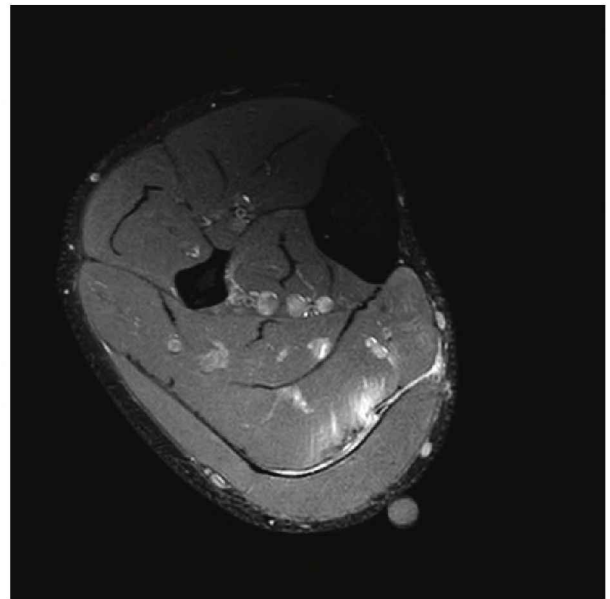
Clinical assessment

Patients with acute soleus strains may present with sudden onset pain, or alternatively with a history of increasing calf tightness over a period of days or weeks. Often walking and jogging are more painful than sprinting. The medial third of the fibres of the soleus and its aponeurosis are prone to becoming hard and inflexible, particularly in athletes with excessive subtalar pronation. This focal tissue can be more susceptible to strain, especially at its junction with adjacent 'normal' tissue. Therefore, careful assessment by palpation as well as assessment of foot biomechanics is essential.

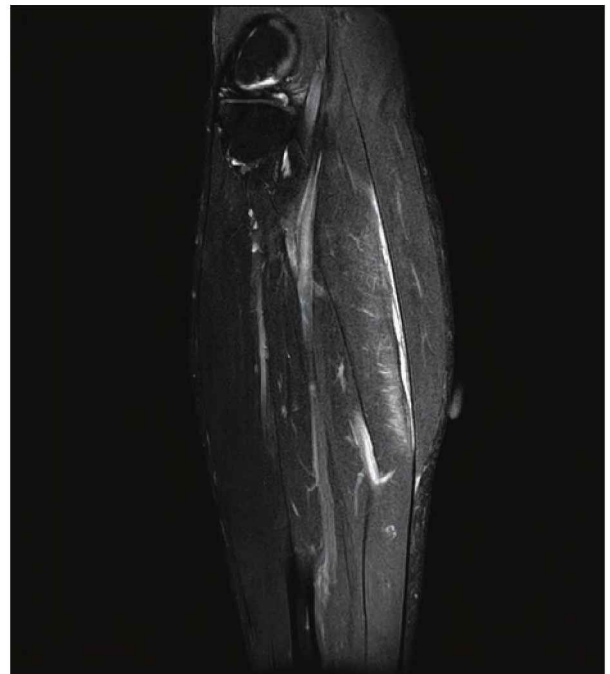
Examination often reveals tenderness deep to the gastrocnemius, usually in the medial aspect of the soleus muscle. Often, both the soleus stretch (Fig. 39.2e) and resisted soleus contraction reproduce the athlete's pain. This can be differentiated from the stretch and contraction that provoke pain in gastrocnemius strains (Fig. 39.2d).

Management

Treatment of soleus muscle strains is similar to gastrocnemius strains above and rehabilitation should target both muscles. It is important to note that activation of the soleus requires the knee to be partially flexed (lunge position). Soft tissue therapy is often directed at the site of the lesion as well as tissue proximally and distally (Fig. 39.8). Strengthening exercises are performed with the knee flexed, initially in a seated position (Fig. 39.9). If foot biomechanics are assessed to be a predisposing factor to injury, orthoses may be required.



(a)



(b)

Figure 39.6 MRI of a 24-year-old running athlete presenting with sudden onset of calf symptoms (a) Axial fat-suppressed image shows epimysial tear of the superficial covering of medial soleus with tearing of the myofibrils from the connective tissue surface. The medial gastrocnemius epimysial surface is intact. Blood fluid products separate the two discrete layers (b) Sagittal fat-suppressed image



(a)



(b)

Figure 39.7 Muscletendinous strain involving the lateral aponeurosis with damage to the aponeurosis itself (a) Sagittal MR image shows disruption of the intramuscular aponeurosis (arrow) of the right lateral soleus muscle with retraction of the aponeurosis to create a wavy appearance. Oedema and blood fluid products track into the torn muscle fibres (b) Axial fat-suppressed image shows a discrete gap at the site where the aponeurosis has torn (arrow). The aponeurosis of the medial soleus muscle is intact



Figure 39.8 Soft tissue therapy—sustained myofascial tension to the soleus muscle

PRACTICE PEARL

When the soleus strain involves a lesion to the soleal aponeuroses (Fig. 39.7), rehabilitation time frames will be extended and management should be based on tendon treatment principles (see Chapter 4).

Accessory soleus

The soleus accessory muscle is a relatively rare anatomical variant (0.7–5.5%) and is bilateral in 10% of cases. It appears as a soft tissue mass bulging medially between the distal part of the tibia and the Achilles tendon, and may be mistaken for a tumour or an inflammatory lesion.¹⁹ Athletes may have a fullness on inspection of their Achilles region from behind.

It may be asymptomatic (25%), or associated with chronic exertional compartment syndrome or posterior tibial nerve compression (tarsal tunnel syndrome). Athletes can present with ankle pain and soft tissue swelling.^{20, 21}

Presentation is usually in adolescence during periods of rapid muscle hypertrophy. Rapid muscle hypertrophy can lead to compartment-type syndrome, posterior tibial nerve compression, tendinopathy and/or partial accessory muscle strain.^{20, 21}

Plain radiography generally shows soft tissue swelling between the deep compartment and the Achilles tendon, which obscures or obliterates Kager's triangle on the



Figure 39.9 Strengthening exercises for the soleus muscle—calf raise with bent knee. This can be made more difficult by adding weights or having the patient drop down over the end of a step

lateral radiograph of the ankle. The appearance of normal muscle on MRI allows the clinician to distinguish it from both abnormal muscle and soft tissue tumours.^{20, 22} Additionally, intrafascial fluid collection and perimascular oedema have been associated with symptomatic accessory soleus muscle presentation on MRI.²¹

Asymptomatic cases require no treatment. If pain or discomfort is present, conservative treatment is trialled first. Conservative treatment can include soft tissue therapy, stretching and strengthening of the calf muscles

and improvement of ankle range when limited.²⁰ Surgery (involving either fasciotomy alone or with excision of the accessory muscle) is recommended when conservative measures fail.^{19, 20}

CLAUDICANT-TYPE CALF PAIN

Another common presentation is the patient who complains of a claudicant-type calf pain with exercise. The differential diagnosis in this patient is a vascular cause (popliteal artery entrapment, atherosclerotic disease, endofibrosis), neuromyofascial causes (referred pain or nerve entrapment) or compartment syndrome (deep posterior or superficial).

Vascular causes

Vascular causes of exercise-induced lower limb pain were originally thought to be uncommon but the incidence may be greater than previously recognised, and it is possible that these conditions are under reported.²³

The pain of vascular entrapment is brought on by exertion, like chronic exertional compartment syndrome, but has subtle differences in that it can be present at rest (positional), will typically develop more quickly and (usually) dissipate more quickly.²⁴

Post-exercise examination of the peripheral pulses and arterial bruits is vital and the diagnosis can be confirmed by Doppler ultrasound, ankle-brachial ratios and angiography. It may also be important to perform compartment pressure tests and nerve conduction studies to rule out coexisting conditions.

Popliteal artery entrapment

Popliteal artery entrapment syndrome (PAES) is often not recognised, and misdiagnosed as a compartment syndrome.^{25, 26} The syndrome was first described as a cause of exercise-induced leg pain in 1879. There are two types of popliteal artery entrapment syndrome: anatomical and functional (acquired).

The classically described anatomical (congenital) abnormality is a variation in the anatomical relationship between the popliteal artery as it exits the popliteal

Table 39.3 Popliteal Vascular Entrapment Forum classification for popliteal entrapment syndrome (adapted from di Marzo and Cavallaro²⁹)

Type I	Popliteal artery running medial to the medial head of gastrocnemius
Type II	Medial head of gastrocnemius laterally attached
Type III	Accessory slip of gastrocnemius/fibrous bands arising from medial head of gastrocnemius
Type IV	Popliteal artery passing below popliteus muscle/fibrous bands arising from popliteus
Type V	Primarily venous entrapment
Type VI	Other variants
Type VII	Functional entrapment



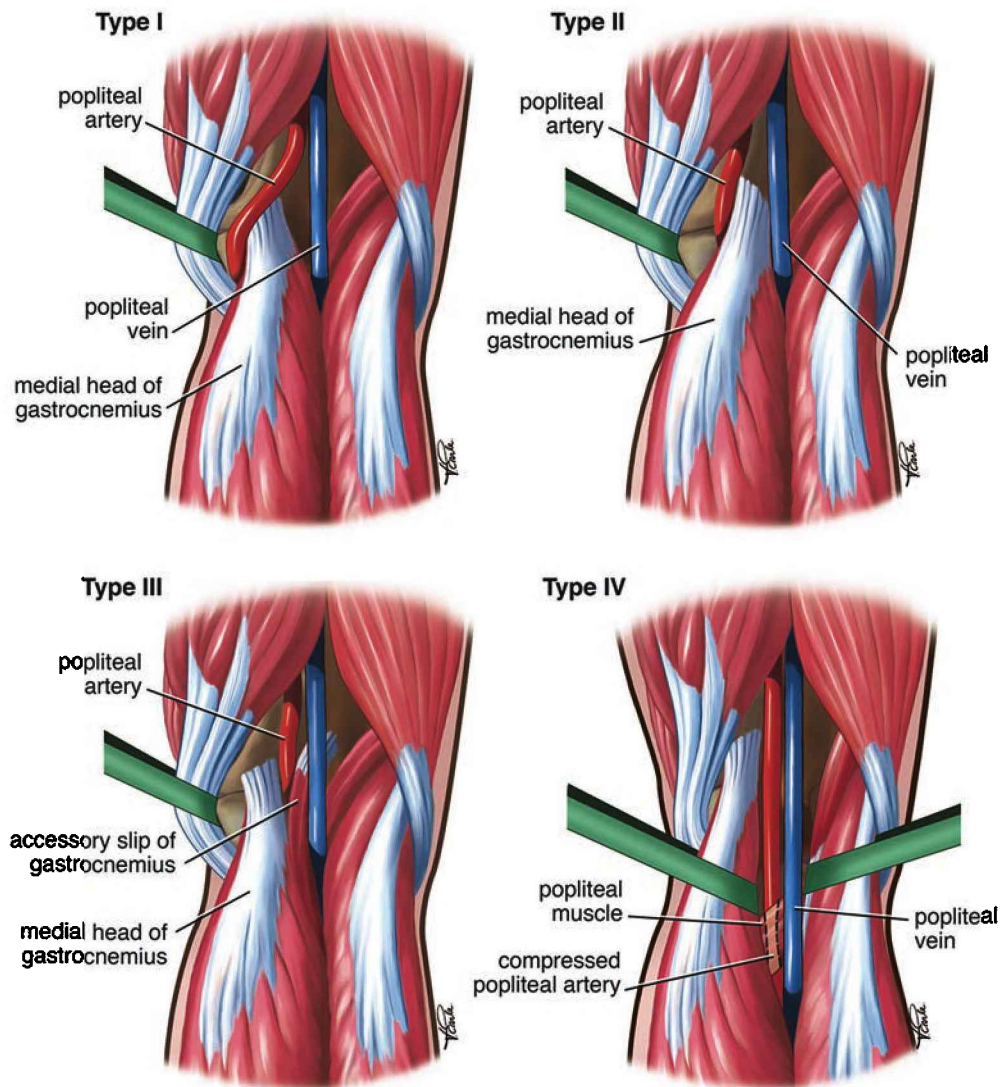


Figure 39.10 Popliteal Vascular Entrapment Forum classification system for PAES.

REDRAWN FROM PILLAI³⁰

fossa and the medial head of the gastrocnemius muscle. It is quite rare, with an estimated prevalence of between 0.6% and 3.5%. Popliteal artery entrapment syndrome can be classified into seven categories according to the anatomical relationship between the popliteal artery and the adjacent muscular or fibrous bands in the popliteal fossa (Table 39.3 and Fig. 39.10).²⁶⁻²⁸

The most common abnormality describes an abnormal medial head of the gastrocnemius muscle, the accessory part of which passes behind the popliteal artery. Other observed abnormalities include a tendinous slip arising from the medial head of the muscle, an abnormal plantaris muscle and multiple muscle abnormalities involving the lateral and medial heads of the gastrocnemius and the plantaris. Rarer anatomical variations of popliteal artery entrapment

syndrome include entrapment of the artery at the level of the adductor hiatus and an isolated entrapment of the anterior tibial artery as it passes through the interosseous membrane.

The term 'functional' or 'acquired' PAES was first described by Rignault et al. in 1985,³¹ and describes a situation where no anatomical abnormality is visible at surgical exploration. Studies have shown that the medial head of gastrocnemius typically 'rests' in a lateral or intercondylar position, leading to crowding of the popliteal vessels and occlusion during muscle contraction (Fig. 39.11).³⁰ This syndrome is commonly seen in healthy young athletes and military recruits with hypertrophied gastrocnemius muscles.^{25, 27}

Clinically, athletes present with claudication-type calf or anterior aspect of leg pain, which can be bilateral. The

Differentiating artery entrapment from chronic compartment syndrome

With artery entrapment, cessation of exercise brings about rapid relief from the pain. This compares with the classic pain pattern of compartment syndrome, which is related to the volume of exercise and tends to settle over a period of around half an hour after exercise. If exercise is attempted on consecutive days, the

pain from compartment syndrome is often more severe on the second day. The pain from PAES is unaffected by exercise on the previous day. The pain can be paradoxically more severe on walking than on running. This is believed to be due to the more prolonged contraction of the gastrocnemius muscle while walking.

B

pain is brought on by exercise, and the severity of the symptoms is related to the intensity of exercise. Signs of ischaemia in the lower limb are uncommon.³²

Examination of the distal pulses (dorsal pedis and posterior tibial) may be unremarkable at rest. Assessing the pulses while the athlete actively plantarflexes or passively dorsiflexes the ankle may reveal a popliteal artery bruit, but this may be unreliable.²⁷ Examining the athlete immediately post-exercise is important in making the diagnosis. Immediately post-exercise, a popliteal artery bruit may be heard and the peripheral pulses will be either weak, decreased or absent.

Ankle-brachial indices are useful for the more rare underlying vascular abnormalities, but not for functional PAES.³³ Doppler ultrasound is the initial investigation of choice, as it allows real-time assessment and can be performed during provocative manoeuvres. Angiography can rule out differential diagnoses such as aneurysms and emboli, and can be performed while the athlete actively plantarflexes, and actively and passively dorsiflexes the ankle.^{27, 28}

MRI and MR angiography may be more useful with multiplanar views, non-ionising radiation and high soft

tissue contrast, and also allow much greater visualisation of the surrounding anatomy.²⁷

There is some suggestion that the presence of chronic entrapment of the popliteal artery can lead to endothelial damage, which may lead to accelerated atherosclerotic disease in later life, especially in cases of anatomical PAES, where early surgical treatment is recommended.³⁴ In cases of functional PAES, the treatment is less clear. A pilot study on the use of ultrasound guided Botox[®] at the site of entrapment is showing promising initial results but requires further research.

Atherosclerotic vessel disease

Atherosclerotic vessel disease classically affects middle-aged sedentary patients. However, some athletes, particularly in the veteran or masters class, fall into the category of middle-aged or elderly and possess risk factors that predispose them to atherosclerosis.

Pain may be felt in the thigh or the calf and is typically claudicant. With progression of the disease, the intensity of exercise needed to produce symptoms decreases. At rest, the peripheral pulses may be difficult to palpate or

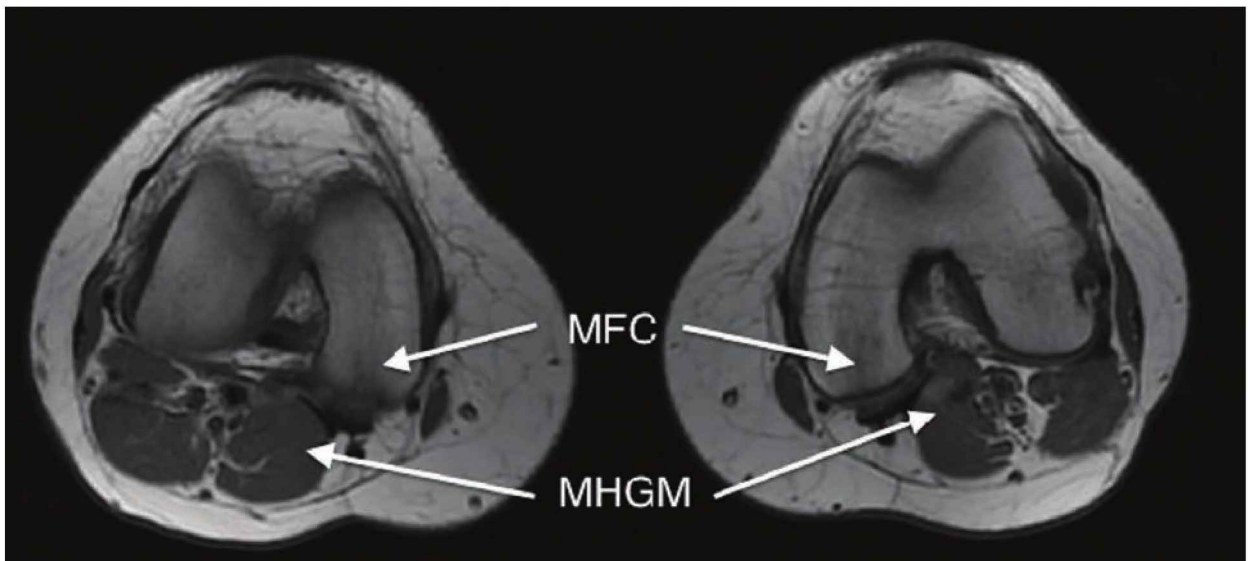


Figure 39.11 Axial MRI images of the popliteal region of both knees. The medial head of gastrocnemius muscle (MHGM) is seen to arise from a more lateral position off the posterior aspect of the medial femoral condyle (MFC) leading to ‘crowding’ of the popliteal vessels

absent. An arterial bruit may also be heard at rest. The presence of such bruits may be enhanced by examining the patient post-exercise.

The gold standard diagnostic test is Doppler ultrasound, although pre- and post-exercise ankle-brachial ratios can be used as a non-invasive screening test. Angiography can confirm the diagnosis.

Surgical treatments include angiographic balloon catheter dilatation or stenting, open endarterectomy or bypass surgery. Bypass surgery is the most commonly used surgical technique for atherosclerotic vessel disease and its success depends on the extent of the disease and on the viability of the smaller distal vessels.

Endofibrosis

Endofibrotic disease can cause exercise-related calf pain, although more commonly the pain is felt in the thigh. Typically the lesion occurs in the proximal external iliac artery, but may extend distally towards the origin of the femoral artery beneath the inguinal ligament. It is bilateral in 15% of cases.

External iliac artery endofibrosis has been described in endurance athletes and professional cyclists, and causes exercise-related thigh or calf pain that is related to the intensity of cycling.^{35, 36} The pain is therefore most commonly felt while racing, climbing a hill or riding into a strong wind. The pain is typically relieved rapidly by a drop in the intensity of exercise.

In cyclists, it is postulated that the cycling position may cause repetitive stress and folding of the artery during high flow rates associated with extreme exercise. As a result, micro-traumatic lesions lead to thickening of arterial walls.³⁶ Psoas hypertrophy may also have a direct effect on the stresses around the external iliac artery.

Examination at rest may reveal a positional bruit heard over the femoral artery with the hip held in flexion. The diagnosis is made clinically by examining the patient immediately post-exercise, detecting a bruit over the femoral artery, and weak or absent distal pulses. Pre- and post-exercise ankle-brachial ratios screen for the diagnosis, which is anatomically confirmed with angiography. Arterial ultrasound or echography can also be useful in visualising the endofibrotic lesion.

Non-surgical techniques for the treatment of external iliac artery endofibrosis include angioplastic balloon catheter dilatation and stenting, which can be planned for and performed at the time of angiography. Surgical techniques include bypass surgery and open endarterectomy. In the short term, use of these techniques has made a return to top-level cycling and triathlon possible in patients.³⁷ However, long-term follow-up of such patients has not been carried out.

The natural history of this pathology is not certain. However, Abraham et al. suggest that it is non-progressive once high-level sport is ceased.³⁸ They advise that an athlete who is reducing his or her sporting level, and who is asymptomatic with activities of daily living and submaximal, be managed conservatively but followed up regularly.

LESS COMMON CAUSES

Neuromyofascial causes

The neural component of calf pain can be assessed with the use of the slump test (Fig. 39.2h). This may reproduce the patient's calf pain. If pain is relieved by cervical extension, neural structures may be contributing to the patient's pain.

The joints of the lumbar spine may occasionally refer pain to the calf. This should be suspected clinically if the pain is somewhat variable in location or if there is a history of recurrent 'calf strains'.

Another source of referred pain is myofascial trigger points in the gluteal muscles. Myofascial pain may present as an episode of sudden sharp pain and mimic a calf strain, as pain of more gradual onset accompanied by tightness, or as muscle cramps. Treatment consists of ischaemic pressure or dry needling to the trigger points followed by muscle and neural stretching.

The knee joint may also refer pain to the calf (Chapter 35). This may be due to a Baker's cyst, or a posterior cruciate ligament injury. Bleeding may track down into the calf from a strain of the posterior capsule of the knee joint or a popliteus muscle injury.

Nerve entrapments

Nerve entrapments around the calf include:

- tibial nerve entrapment secondary to a Baker's cyst (rare)
- popliteal artery aneurysm or ganglion resulting in ankle inverter and toe flexor weakness and paraesthesia to the sole of the foot
- sural nerve entrapment, which may result from compression (ski boots, casts), mass lesions, trauma or thrombophlebitis causing pain and paraesthesia in the lateral heel and foot.

The sural nerve may be compressed in a fibrous arch that thickens the superficial sural aponeurosis around the opening through which the nerve passes. Intense physical training may lead to an increase in the sural muscle mass, which in turn compromises the sural nerve in its trajectory through the unyielding and inextensible superficial sural aponeurosis. The nerve may also become trapped in scar tissue.³⁹

Investigations can include nerve conduction studies +/- MRI to detect a space-occupying lesion. Management includes conservative measures such as neural stretches, fascial massage or corticosteroid injections, and failing this, consideration of surgical exploration and/or neurolysis.

Superficial compartment syndrome

Patients with superficial posterior compartment syndrome, the least common of the lower leg compartment syndromes, present with calf pain. The superficial compartment contains the gastrocnemius and soleus muscles, which are enclosed in a fascial sheath. Symptoms are similar to those of the other compartment syndromes with pain aggravated by activity and relieved by rest. An elevated compartment pressure confirms the diagnosis during and after exercise. Treatment consists of soft tissue therapy or, if this is unsuccessful, surgery.

Patients with either deep posterior compartment syndrome or stress fracture involving the posterior cortex

of the tibia may present with calf pain instead of, or as well as, shin pain (Chapter 38).

CONDITIONS NOT TO BE MISSED

Deep venous thrombosis occurs occasionally in association with calf injuries. The post-injury combination of lack of movement, disuse of the muscle pump and the compressive effect of swelling may all lead to venous dilatation, pooling and a decrease in the velocity of blood flow. Certainly athletes who sustain a calf muscle injury should avoid travelling on long flights in the days after injury.

Deep venous thrombosis is a rare, but potentially fatal, risk after arthroscopy.⁴⁰ The diagnosis should be suspected when the patient has constant calf pain, tenderness, increased temperature and swelling. Homan's sign (passive dorsiflexion) (Fig. 39.2c) may be positive. Homan's test is taught to all medical students but has poor diagnostic utility. The best test for deep vein thrombosis is Wells' clinical prediction rule.⁴¹ The presence of deep venous thrombosis may be confirmed by Doppler scan and venography.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

